

Saichand Vemuri

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Location: San Francisco, CA, USA

SUMMARY

Full Stack Developer with 5 years of experience building scalable, high-performance applications in finance and healthcare. Skilled in **Python, Django, FastApi, GraphQL, React, REST APIs, Kafka, Redis, Microservices, SQL/NoSQL, and CI/CD**. Experienced in solving complex engineering challenges, delivering production-ready solutions, and collaborating with cross-functional teams to optimize workflows and improve user experience.

SKILLS

- Frontend:** React, Angular, TypeScript, JavaScript ES6, Redux, Tailwind, Next.js
- Backend:** Python, FastAPI, Django, Django REST Framework, Java, Spring Boot, Node, Express
- Infrastructure / Cloud:** AWS Lambda, API Gateway, ECS, Fargate, S3, DynamoDB, RDS/MySQL, CloudWatch, Docker, Redis, Kafka
- Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit
- Machine Learning & LLMs:** PyTorch, HuggingFace, RAG, NLP, Bedrock, Sagemaker
- Tools:** Git, Jira, Figma, Sentry, Prometheus, Grafana
- Practices:** System Design, API Architecture, Microservices, Performance Optimization, CI/CD, Agile

PROFESSIONAL EXPERIENCE

Full Stack Developer, CVS Health, USA

September 2023 – Present

- Developed scalable **React, TypeScript, JavaScript** applications using **Redux** and React **hooks** for reusable, maintainable components. Optimized frontend workflows and performance, reducing load times by 40% and improving overall user experience.
- Developed **Python/FastAPI pipelines** with NLP and custom parsers, increasing claim-processing speed by 60% and reducing manual review hours.
- Used PyTorch with **AWS SageMaker** to train and deploy **ML models**, applying regression, classification, and anomaly detection to identify high-risk patients and operational outliers, improving accuracy by **~18%**.
- Worked and trained a deep learning-based vision model in **PyTorch** for retail shelf analysis, using a CNN backbone with custom detection and classification heads to identify products, detect out-of-stock items, and flag misplaced products from shelf images.
- Built **AI-powered agents** using **domain-tuned LLMs** (OpenAI GPT, Meta llama) to summarize medical histories, flag inconsistencies, and identify recoverable claims, improving case preparation efficiency by **40%**.
- Improved efficiency of AI query handling by building **Kafka**-based streaming pipelines, **Redis caching**, and **SQL**, enabling the system to process thousands of concurrent user queries while maintaining fast, reliable, and consistent responses.
- Designed and deployed **AWS solutions** (Lambda, API Gateway, S3, ECS, Secrets Manager, ElastiCache) with monitoring via **CloudWatch, Prometheus, Grafana**, reducing troubleshooting time by **40%**.
- Developed **Python** scripts to automate data processing, system monitoring, and workflow tasks, improving operational efficiency, reducing manual effort, and enabling reliable, repeatable processes across Linux environments.
- Built and owned **MLOps** pipelines with **CI/CD**, orchestrating end-to-end ML workflows in **AWS SageMaker** while monitoring performance, scalability, and cost under high-volume production workloads.

Associate Software Engineer, Ahex Technologies Private Limited, USA

January 2020 – May 2023

- Developed scalable **Django** microservices with **RESTful** and **GraphQL APIs** to manage trading data, user accounts, and transaction histories, enabling high-throughput, reliable backend services for real-time trading operations.
- Implemented automated file handling pipelines for trading and insurance documents using **Django** scheduled tasks with Celery, reducing error files and ensuring high-availability operations.
- Designed and optimized Python-based network services for data exchange over **TCP/IP and HTTP protocols**, improving latency and reliability in trading and market data feeds.
- Built Python worker components to process message queues **Kafka** and transform log data in **Unix/Linux** environments, significantly reducing processing time for large-scale transactions.
- Applied **Django** REST Framework authentication, **OAuth 2.0**, and **LDAP integration** to secure backend services, ensuring robust authorization for trading platforms and achieving high test coverage (>95%) for API endpoints.
- Actively contributed to **Agile** ceremonies (planning, estimation, standups, retrospectives) and maintained **Git/GitHub** version control, streamlining collaboration and deployment.
- Implemented **CI/CD pipelines** with GitHub Actions to build, test, and deploy Python/Django services to **Kubernetes clusters** using Helm, improving deployment speed and reliability.

PROJECTS

Smart Diagram Question Answering

- Built and trained deep learning models in **PyTorch** to analyze images and diagrams, achieving accurate automated **visual reasoning**.
- Integrated **CNN** based vision encoders with transformer-based language models to answer natural language questions from visual inputs.
- Designed a complete **PyTorch** pipeline for image preprocessing, model training, inference, and evaluation, demonstrating practical multimodal AI applications.

EDUCATION

- Master of Science in Computer Science
- University Of Central Missouri, USA

CERTIFICATIONS & ACHIEVEMENTS

- Certified in System Design LLD and HLD, demonstrating expertise in architecting scalable, efficient systems with a strong focus on design principles, performance optimization, and real-world application.
- Collaborated closely with cross-functional teams including product managers, data scientists, and operations stakeholders to translate retail business requirements into scalable vision-based deep learning solutions.